MITOCHONDRIA · ASTRONAUT HEALTH INTELLIGENCE

Health Intelligence Report

Preliminary automated assessment — for Ground Medical review. Not a physician's diagnosis.

ASTRONAUT	MISSION	ASSESSMENT PERIOD	EXAMINATIONS WITH DATA
Alex Morgan — Synthetic Astronaut	MITO_DEMO_2026	2026-11-01 to 2026-11-30	3, 6, 9, 12, 16, 20, 24, 28 Nov 2026 (8 dates)

SYNTHETIC DEMONSTRATION DATA — not real astronaut data. Age, sex, symptoms, medications and medical history are not recorded and are not assumed.

Most important findings

MONITORING OR FURTHER EVALUATION

Clinical class NEUTRAL

Possible electrolyte disturbance (sodium / potassium)

Sodium: 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 146 mmol/L on 2026-11-20 (intermediate maximum), then fell again; first to last 0 mmol/L (0%). Potassium: 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days).

Why this priority: Significant change (trend review): Potassium; no RED band or diagnostic/action threshold supports high priority.

Next: Repeat sodium and potassium promptly under Ground Medical supervision with careful sample handling; the timing is a clinical decision.

MONITORING OR FURTHER EVALUATION

Clinical class NEUTRAL

Possible inflammatory response

hsCRP (inflammation): 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 28 mg/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.9 mg/L (+75%).

Why this priority: Significant change (trend review): hsCRP (inflammation), IL-6 (inflammatory signal); no RED band or diagnostic/action threshold supports high priority.

Next: Repeat the inflammatory markers and examine for sources of infection or inflammation; the CDC/AHA statement recommends repeating hs-CRP above 10 mg/L.

MONITORING OR FURTHER EVALUATION

Clinical class NEUTRAL DATA CHECK

Possible disturbance of blood-sugar regulation (dysglycemia)

HbA1c (3-month average blood sugar): 5.8 → 5.9 → 9.1 → 5.9 → 5.8 → 5.8 → 5.8 → 5.8 % from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 9.1 % on 2026-11-09 (intermediate maximum), then fell again; first to last 0 % (0%).

Why this priority: Significant change (trend review): Fasting glucose; no RED band or diagnostic/action threshold supports high priority. Verify first: HbA1c (3-month average blood sugar).

Next: Confirm any criterion-level result with a second abnormal test (same test at a later time point, or a different test), as required by the ADA Standards of Care.

MONITORING OR FURTHER EVALUATION

3 further findings

Monitoring or further evaluation: Raised blood triglycerides (possible lipid-metabolism concern); Change in bone-turnover markers (possible bone-health concern); Reduced heart-rate variability (possible recovery or autonomic strain).

Next: See the Preliminary Medical Assessment.

RED	High priority medical review; not a diagnosis	AMBER	Monitoring or further evaluation	DATA CHECK	Purple — verify the measurement first
GREEN	Within applicable reference criteria	NEUTRAL	Insufficient classification evidence	TREND REVIEW	Significant change between examinations; not a priority

Clinical classes: 0 Researcher-approved criteria for this mission. Panel colour = warning priority, separate from each result's clinical class. General-population criteria are not NASA flight limits.

Priority next steps

1. Repeat sodium and potassium promptly under Ground Medical supervision with careful sample handling; the timing is a clinical decision.
2. Repeat the inflammatory markers and examine for sources of infection or inflammation; the CDC/AHA statement recommends repeating hs-CRP above 10 mg/L.

Executive health overview

8 examination date(s) with 248 observation(s) in 31 biomarker series. 7 body system(s) show flagged or classified results; 17 series show a significant change between examinations (trend review) and 1 require measurement verification first. Medical-review priority: 0 finding(s) high priority, 6 monitoring, 0 data check.

1 · What is happening?

6 possible concern area(s) for Ground Medical review: possible electrolyte disturbance; possible inflammatory response; possible disturbance of blood-sugar regulation; raised blood triglycerides; change in bone-turnover markers; reduced heart-rate variability. 6 concern areas changed in the same period across Cardiovascular Health, Bone Health: possible electrolyte disturbance; possible inflammatory response; possible disturbance of blood-sugar regulation; raised blood triglycerides; change in bone-turnover markers; reduced heart-rate variability. Their co-occurrence is reported for joint review; it does not show that one caused another. Shared context such as fluid shifts, sleep, stress, illness or sample handling should be checked first.

2 · Which results are abnormal or concerning?

No result is clinically classified (no applicable criterion). Trend review — significant change between examinations (not a medical-review priority or a classification): Potassium, Fasting glucose, HbA1c (3-month average blood sugar), Triglycerides, hsCRP (inflammation), IL-10 (anti-inflammatory signal), IL-6 (inflammatory signal), TNF-α, D-dimer, Fibrinogen, Lactate, Resting heart rate, CTX (bone breakdown), Osteocalcin (bone formation), Heart-rate variability (RMSSD), Respiratory rate, White blood cell count.

3 · What increased or decreased?

Increased: Fasting glucose 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL (+2.1 %) (intermediate peak 172); LDL cholesterol 122 → 127 → 134 → 145 → 143 → 137 → 131 → 125 mg/dL (+2.5 %) (intermediate peak 145); Triglycerides 155 → 165 → 230 → 330 → 288 → 216 → 184 → 160 mg/dL (+3.2 %) (intermediate peak 330); hsCRP (inflammation) 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L (+75.0 %) (intermediate peak 28); IL-10 (anti-inflammatory signal) 4 → 5 → 8 → 15 → 13 → 9 → 6 → 5 pg/mL (+25.0 %) (intermediate peak 15); IL-6 (inflammatory signal) 5.8 → 12 → 32 → 74 → 55 → 29 → 15 → 8.1 pg/mL (+39.7 %) (intermediate peak 74); TNF-α 8 → 10 → 15 → 24 → 20 → 15 → 11 → 9 pg/mL (+12.5 %) (intermediate peak 24); D-dimer 0.32 → 0.41 → 0.82 → 2.8 → 2.1 → 1.2 → 0.69 → 0.39 mg/L FEU (+21.9 %) (intermediate peak 2.8); Fibrinogen 295 → 320 → 390 → 520 → 485 → 420 → 350 → 305 mg/dL (+3.4 %) (intermediate peak 520); Lactate 1.1 → 1.3 → 1.8 → 4.2 → 3.0 → 1.9 → 1.4 → 1.2 mmol/L (+9.1 %) (intermediate …

4 · Which changes are biologically meaningful?

Plausible and important if confirmed: Potassium, Fasting glucose, Triglycerides, hsCRP (inflammation), IL-10 (anti-inflammatory signal), IL-6 (inflammatory signal), TNF-α, D-dimer, Fibrinogen, Lactate, Resting heart rate, CTX (bone breakdown), Osteocalcin (bone formation), Heart-rate variability (RMSSD), Respiratory rate, White blood cell count. Verify the measurement first: HbA1c (3-month average blood sugar).

5 · Which body systems are affected?

Fluids & electrolytes; Glucose & lipids; Inflammation & immunity; Blood & clotting; Heart & circulation; Bone & mineral; Sleep & recovery

6 · How are the biomarkers related?

Inflammation – Lipids: raises triglycerides, alters HDL (established); Red cells – Glucose / HbA1c: altered red-cell turnover distorts HbA1c (established); Inflammation – Recovery / autonomic: co-occurs (not causal); Glucose / HbA1c – Lipids: co-occurs (insulin resistance is a hypothesis). Co-occurrence does not show cause.

7 · Which conditions could explain the findings?

Possible electrolyte disturbance (sodium / potassium); Possible inflammatory response; Possible disturbance of blood-sugar regulation (dysglycemia); Raised blood triglycerides (possible lipid-metabolism concern); Change in bone-turnover markers (possible bone-health concern); Reduced heart-rate variability (possible recovery or autonomic strain) All are preliminary and require clinical evaluation.

8 · What supports or contradicts them?

Possible electrolyte disturbance: 2 supporting, 2 contradicting item(s); Possible inflammatory response: 2 supporting, 2 contradicting item(s); Possible disturbance of blood-sugar regulation: 1 supporting, 3 contradicting item(s); Raised blood triglycerides: 3 supporting, 3 contradicting item(s); Change in bone-turnover markers: 4 supporting, 3 contradicting item(s); Reduced heart-rate variability: 1 supporting, 1 contradicting item(s). Most often missing: Kidney function (creatinine / eGFR); Urine sodium and potassium, urine volume; Sample handling and hemolysis index; Fluid intake and urine output.

9 · What should the astronaut do?

Do not change fluid, salt or potassium intake before Ground Medical review. Report any symptoms. Maintain the prescribed aerobic and resistance exercise; muscle activity increases glucose uptake and insulin sensitivity. Regular aerobic exercise increases the clearance of triglycerides from the blood; maintain the prescribed exercise programme.

10 · What should Ground Medical confirm?

Ground Medical should review calcium and vitamin D adequacy; any decision about supplementation belongs to Ground Medical. Do not change fluid or salt intake on the basis of these results until Ground Medical has reviewed them; the appropriate direction depends on the cause. Request a repeat HbA1c from a new sample and a fasting plasma glucose with documented fasting time, so that a criterion-level result can be confirmed or excluded. Repeat a fasting lipid panel with documented fasting time to establish whether triglycerides remain raised.

Body systems at a glance

Fluids & electrolytes
TREND REVIEW

Potassium 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L; Sodium 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L

Glucose & lipids
TREND REVIEW DATA CHECK

Fasting glucose 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL; HbA1c (3-month average blood sugar) 5.8 → 5.9 → 9.1 → 5.9 → 5.8 → 5.8 → 5.8 → 5.8 %; Triglycerides 155 → 165 → 230 → 330 …

Inflammation & immunity
TREND REVIEW

hsCRP (inflammation) 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L; IL-10 (anti-inflammatory signal) 4 → 5 → 8 → 15 → 13 → 9 → 6 → 5 pg/mL; IL-6 (inflammatory signal) 5.8 → 12 → 32 → …

Blood & clotting
TREND REVIEW

D-dimer 0.32 → 0.41 → 0.82 → 2.8 → 2.1 → 1.2 → 0.69 → 0.39 mg/L FEU; Fibrinogen 295 → 320 → 390 → 520 → 485 → 420 → 350 → 305 mg/dL; aPTT (clotting time) 29 → 30 → 31 → 34 → 32 …

Heart & circulation
TREND REVIEW

Lactate 1.1 → 1.3 → 1.8 → 4.2 → 3.0 → 1.9 → 1.4 → 1.2 mmol/L; Resting heart rate 64 → 69 → 77 → 108 → 98 → 83 → 72 → 66 bpm; Diastolic blood pressure 76 → 73 → 69 → 54 → 58 → 65 → 72 → 75 …

Bone & mineral
TREND REVIEW

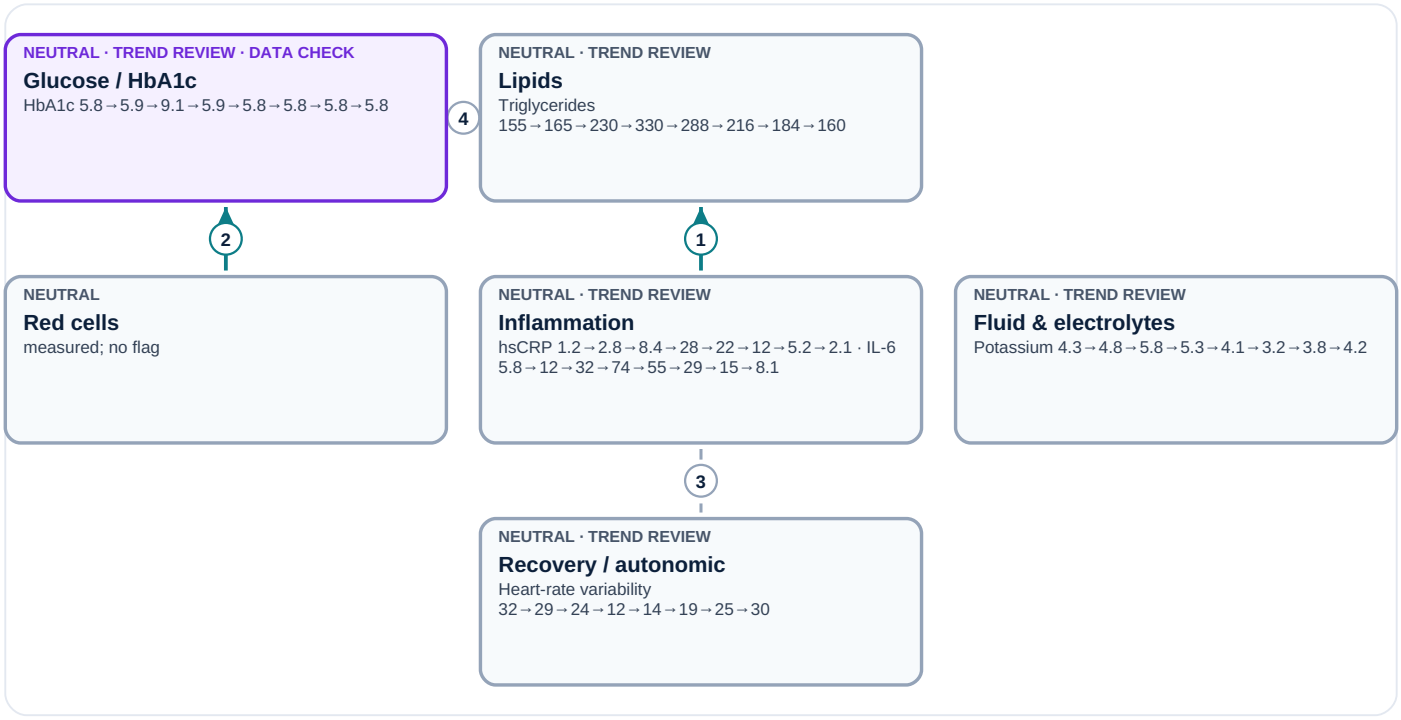
CTX (bone breakdown) 0.32 → 0.35 → 0.41 → 0.73 → 0.66 → 0.55 → 0.45 → 0.36 ng/mL; Osteocalcin (bone formation) 3.5 → 4.1 → 5.2 → 31 → 25 → 18 → 11 → 7.2 ng/mL; P1NP (bone formation) 43 → 45 …

Sleep & recovery
TREND REVIEW

Heart-rate variability (RMSSD) 32 → 29 → 24 → 12 → 14 → 19 → 25 → 30 ms; Sleep efficiency 83 → 78 → 69 → 52 → 56 → 66 → 75 → 81 %

Integrated health assessment

Findings are read as connected patterns. Solid arrows: established biological links (with source). Dashed lines: co-occurrence in this period or a labelled hypothesis — neither shows cause.



1. Inflammation — Lipids. Established: raises triglycerides, alters HDL.
2. Red cells — Glucose / HbA1c. Established: altered red-cell turnover distorts HbA1c.
3. Inflammation — Recovery / autonomic. Co-occurrence: co-occurs (not causal).
4. Glucose / HbA1c — Lipids. Co-occurrence: co-occurs (insulin resistance is a hypothesis).

Possible electrolyte disturbance

Sodium: 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 146 mmol/L on 2026-11-20 (intermediate maximum), then fell again; first to last 0 mmol/L (0%). Potassium: 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 5.8 mmol/L on 2026-11-09 (intermediate maximum), then fell again; first to last -0.1 mmol/L (-2.3%). Sodium and potassium are held within narrow limits because they control fluid distribution and the electrical activity of nerves and heart muscle. Large changes, particularly in potassium, can affect heart rhythm and therefore need prompt medical review rather than waiting for the next routine examination.

Possible mechanisms (not established for this crew member): Changes in fluid intake, fluid loss or fluid shifts (including the headward fluid shift of microgravity). Hormonal regulation by aldosterone and vasopressin acting on the kidneys. Changes in kidney function.

Possible inflammatory response

hsCRP (inflammation): 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 28 mg/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.9 mg/L (+75%). IL-6 (inflammatory signal): 5.8 → 12 → 32 → 74 → 55 → 29 → 15 → 8.1 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 74 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +2.3 pg/mL (+39.7%). Inflammatory markers rise when the immune system responds to infection, injury, strenuous exercise or stress. IL-6 is a short-lived signalling molecule; CRP is produced by the liver in response to IL-6 over the following day. A marked rise calls for a search for its cause.

Possible mechanisms (not established for this crew member): Infection or tissue injury activating the innate immune response. Strenuous or unaccustomed exercise causing muscle micro-injury. Psychological or physiological stress and sleep loss.

Possible disturbance of blood-sugar regulation

HbA1c (3-month average blood sugar): 5.8 → 5.9 → 9.1 → 5.9 → 5.8 → 5.8 → 5.8 → 5.8 % from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 9.1 % on 2026-11-09 (intermediate maximum), then fell again; first to last 0 % (0%). Fasting glucose: 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 172 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +2 mg/dL (+2.1%). HbA1c (glycated hemoglobin) reflects average blood glucose over roughly the previous two to three months, while plasma glucose reflects the moment of sampling. Persistently raised values indicate that glucose regulation by insulin may be impaired, which over time affects blood vessels, nerves, kidneys and the eyes.

Possible mechanisms (not established for this crew member): Reduced insulin sensitivity (insulin resistance), for example with physical inactivity, stress hormones or sleep loss. Reduced insulin secretion by the pancreas. Recent food intake or acute stress raising a single glucose reading.

Raised blood triglycerides

Triglycerides: 155 → 165 → 230 → 330 → 288 → 216 → 184 → 160 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 330 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +5 mg/dL (+3.2%). LDL cholesterol: 122 → 127 → 134 → 145 → 143 → 137 → 131 → 125 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 145 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 mg/dL (+2.5%). HDL cholesterol: 48 → 46 → 43 → 39 → 40 → 42 → 44 → 46 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 39 mg/dL on 2026-11-12 (intermediate minimum), then rose again; first to last -2 mg/dL (-4.2%). Triglycerides are the main form in which the body transports and stores fat. Levels rise for several hours after a meal; persistently raised fasting levels are associated with heart and blood-vessel complications, and very high levels can inflame the pancreas.

Possible mechanisms (not established for this crew member): Recent food intake, particularly sugars, refined carbohydrates or alcohol. Insulin resistance increasing triglyceride production by the liver. Reduced physical activity lowering triglyceride clearance.

Change in bone-turnover markers

Osteocalcin (bone formation): 3.5 → 4.1 → 5.2 → 31 → 25 → 18 → 11 → 7.2 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 31 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3.7 ng/mL (+105.7%). CTX (bone breakdown): 0.32 → 0.35 → 0.41 → 0.73 → 0.66 → 0.55 → 0.45 → 0.36 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 0.73 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +0.04 ng/mL (+12.5%). P1NP (bone formation): 43 → 45 → 49 → 68 → 65 → 57 → 51 → 46 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 68 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 ng/mL (+7%). Vitamin D (25-hydroxy): 12 → 12 → 12 → 12 → 13 → 13 → 14 → 14 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Rose at every examination; first to last +2 ng/mL (+16.7%). Bone is continuously rebuilt: osteoblasts form new bone (formation markers such as osteocalcin and P1NP) while osteoclasts break it down (resorption markers such as CTX). In microgravity, unloading increases resorption, so formation and resorption markers, calcium, parathyroid hormone and vitamin D are interpreted together.

Possible mechanisms (not established for this crew member): Skeletal unloading in microgravity increasing bone resorption. Changes in vitamin D status or calcium supply affecting mineralisation. Changes in exercise loading.

Reduced heart-rate variability

Heart-rate variability (RMSSD): 32 → 29 → 24 → 12 → 14 → 19 → 25 → 30 ms from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 12 ms on 2026-11-12 (intermediate minimum), then rose again; first to last -2 ms (-6.3%). Heart-rate variability (for example RMSSD) reflects the balance of the autonomic nervous system. Lower values often accompany fatigue, stress, sleep loss, illness or heavy training, and can indicate incomplete recovery; they are highly dependent on measurement conditions.

Possible mechanisms (not established for this crew member): Increased sympathetic (stress) activity or reduced parasympathetic activity. Sleep restriction and circadian disruption. Accumulated training load or illness.

Across body systems

6 concern areas changed in the same period across Cardiovascular Health, Bone Health: possible electrolyte disturbance; possible inflammatory response; possible disturbance of blood-sugar regulation; raised blood triglycerides; change in bone-turnover markers; reduced heart-rate variability. Their co-occurrence is reported for joint review; it does not show that one caused another. Shared context such as fluid shifts, sleep, stress, illness or sample handling should be checked first.

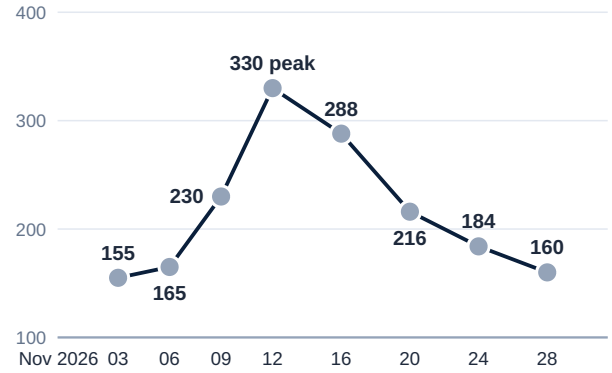
Longitudinal biomarker intelligence

Every chart shows only observed values, on their actual dates and in their original units. Coloured bands appear only where an approved (or, in demonstration mode, a provisional literature) criterion applies. Hollow violet markers and dashed lines: measurement must be verified first. Grey: unclassified.

Triglycerides

mg/dL

MONITORING TREND REVIEW

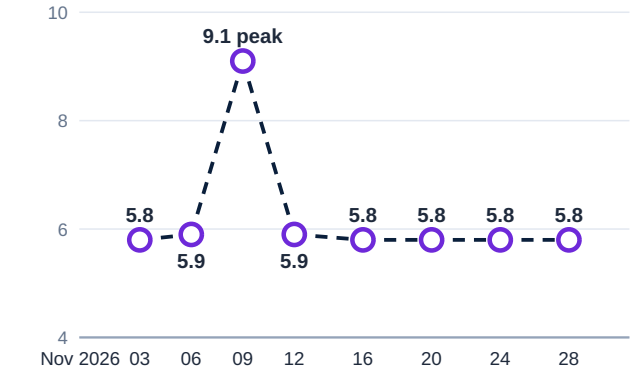


Triglycerides. 155 → 165 → 230 → 330 → 288 → 216 → 184 → 160 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 330 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +5 mg/dL (+3.2%). No approved reference criterion applies, so no reference band is drawn.

HbA1c (3-month average blood sugar)

%

MONITORING TREND REVIEW DATA CHECK

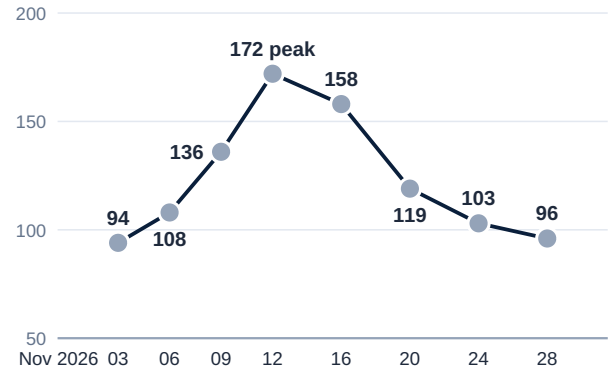


HbA1c (3-month average blood sugar). 5.8 → 5.9 → 9.1 → 5.9 → 5.8 → 5.8 → 5.8 → 5.8 % from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 9.1 % on 2026-11-09 (intermediate maximum), then fell again; first to last 0 % (0%). 5.9 → 9.1 % (+54.2%) from 2026-11-06 to 2026-11-09: a 3-day interval is far shorter than the period HbA1c reflects; measurement consistency must be checked before interpretation.

Fasting glucose

mg/dL

MONITORING TREND REVIEW

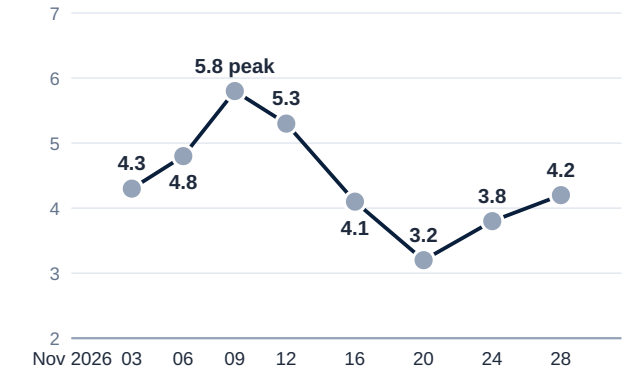


Fasting glucose. 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 172 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +2 mg/dL (+2.1%). No approved reference criterion applies, so no reference band is drawn.

Potassium

mmol/L

MONITORING TREND REVIEW

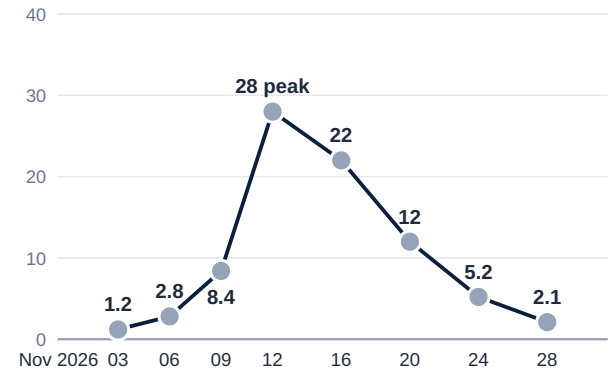


Potassium. 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 5.8 mmol/L on 2026-11-09 (intermediate maximum), then fell again; first to last -0.1 mmol/L (-2.3%). No approved reference criterion applies, so no reference band is drawn.

hsCRP (inflammation)

mg/L

MONITORING TREND REVIEW

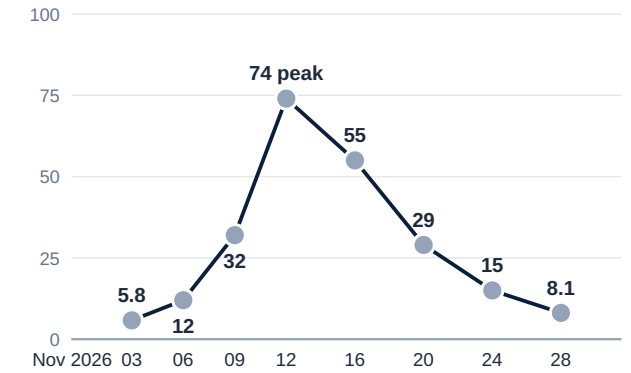


hsCRP (inflammation). 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 28 mg/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.9 mg/L (+75%). No approved reference criterion applies, so no reference band is drawn.

IL-6 (inflammatory signal)

pg/mL

MONITORING TREND REVIEW

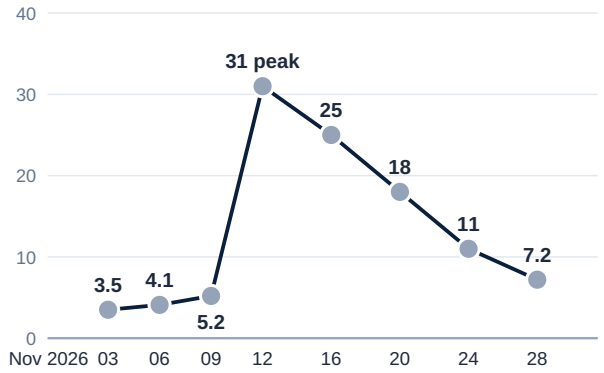


IL-6 (inflammatory signal). 5.8 → 12 → 32 → 74 → 55 → 29 → 15 → 8.1 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 74 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +2.3 pg/mL (+39.7%). No approved reference criterion applies, so no reference band is drawn.

Osteocalcin (bone formation)

ng/mL

MONITORING TREND REVIEW

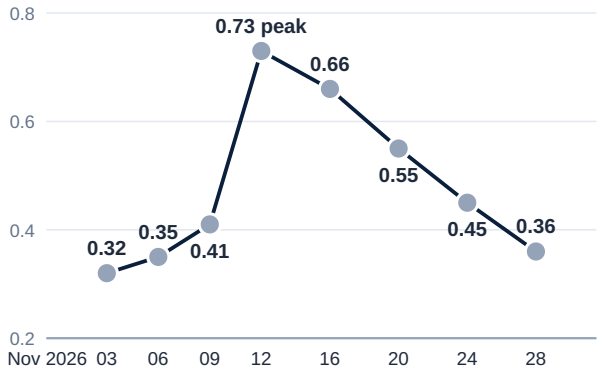


Osteocalcin (bone formation). 3.5 → 4.1 → 5.2 → 31 → 25 → 18 → 11 → 7.2 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 31 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3.7 ng/mL (+105.7%). No approved reference criterion applies, so no reference band is drawn.

CTX (bone breakdown)

ng/mL

MONITORING TREND REVIEW

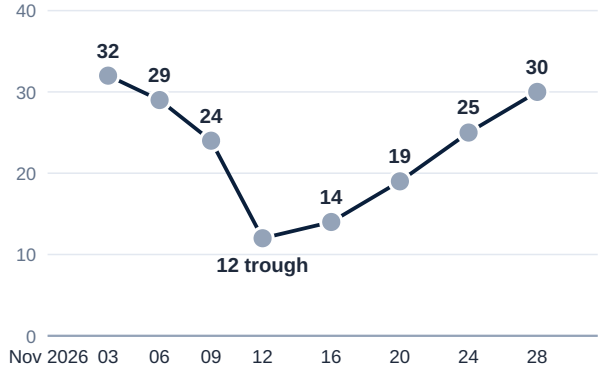


CTX (bone breakdown). 0.32 → 0.35 → 0.41 → 0.73 → 0.66 → 0.55 → 0.45 → 0.36 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 0.73 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +0.04 ng/mL (+12.5%). No approved reference criterion applies, so no reference band is drawn.

Heart-rate variability (RMSSD)

ms

MONITORING TREND REVIEW

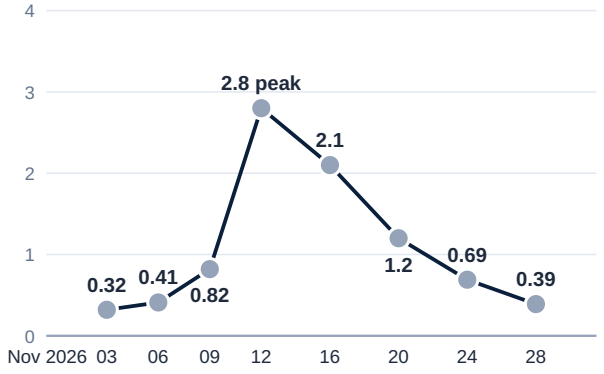


Heart-rate variability (RMSSD). 32 → 29 → 24 → 12 → 14 → 19 → 25 → 30 ms from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 12 ms on 2026-11-12 (intermediate minimum), then rose again; first to last -2 ms (-6.3%). No approved reference criterion applies, so no reference band is drawn.

D-dimer

mg/L FEU

TREND REVIEW

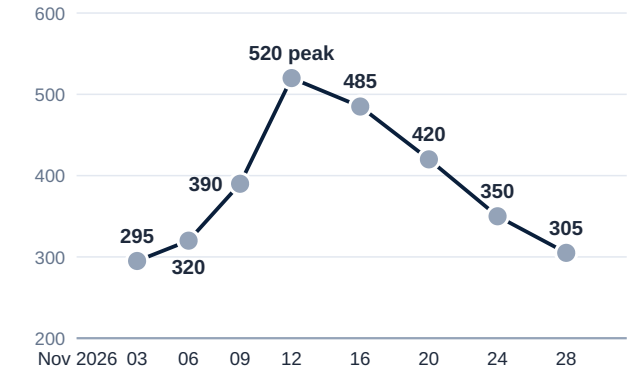


D-dimer. 0.32 → 0.41 → 0.82 → 2.8 → 2.1 → 1.2 → 0.69 → 0.39 mg/L FEU from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 2.8 mg/L FEU on 2026-11-12 (intermediate maximum), then fell again; first to last +0.07 mg/L FEU (+21.9%). No approved reference criterion applies, so no reference band is drawn.

Fibrinogen

mg/dL

TREND REVIEW

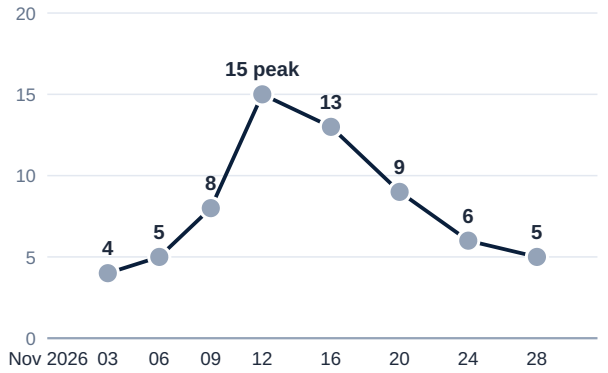


Fibrinogen. 295 → 320 → 390 → 520 → 485 → 420 → 350 → 305 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 520 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +10 mg/dL (+3.4%). No approved reference criterion applies, so no reference band is drawn.

IL-10 (anti-inflammatory signal)

pg/mL

TREND REVIEW

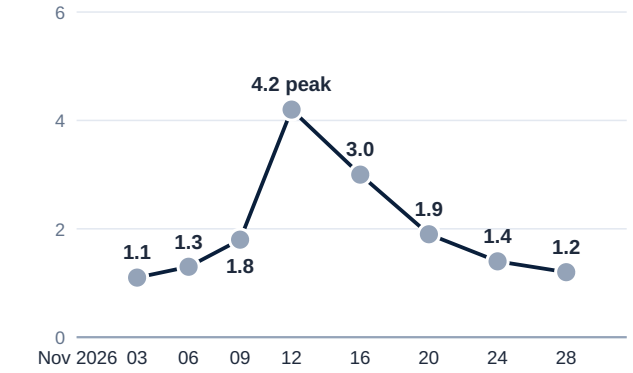


IL-10 (anti-inflammatory signal). 4 → 5 → 8 → 15 → 13 → 9 → 6 → 5 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 15 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +1 pg/mL (+25%). No approved reference criterion applies, so no reference band is drawn.

Lactate

mmol/L

TREND REVIEW

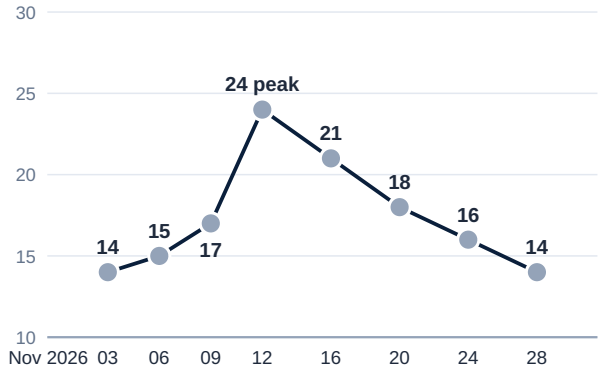


Lactate. 1.1 → 1.3 → 1.8 → 4.2 → 3.0 → 1.9 → 1.4 → 1.2 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 4.2 mmol/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.1 mmol/L (+9.1%). No approved reference criterion applies, so no reference band is drawn.

Respiratory rate

breaths/min

TREND REVIEW

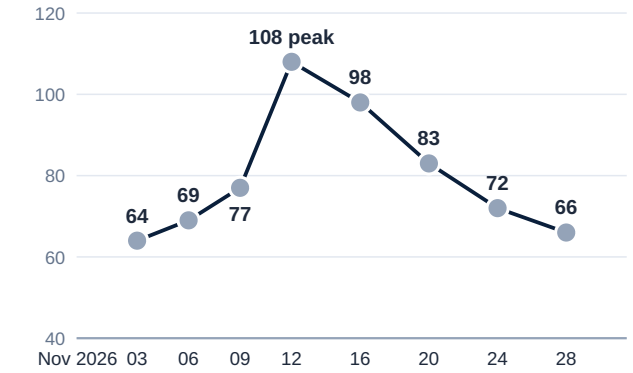


Respiratory rate. 14 → 15 → 17 → 24 → 21 → 18 → 16 → 14 breaths/min from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 24 breaths/min on 2026-11-12 (intermediate maximum), then fell again; first to last 0 breaths/min (0%). No approved reference criterion applies, so no reference band is drawn.

Resting heart rate

bpm

TREND REVIEW

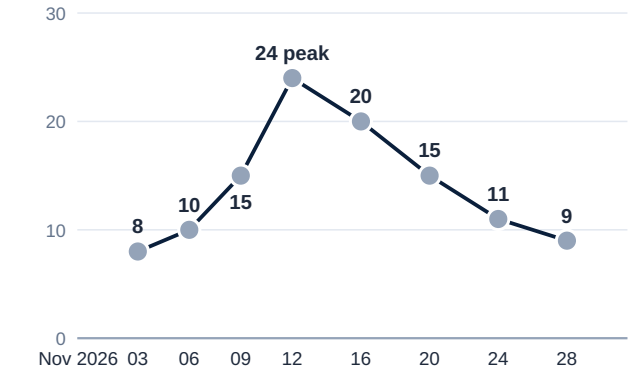


Resting heart rate. 64 → 69 → 77 → 108 → 98 → 83 → 72 → 66 bpm from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 108 bpm on 2026-11-12 (intermediate maximum), then fell again; first to last +2 bpm (+3.1%). No approved reference criterion applies, so no reference band is drawn.

TNF-α

pg/mL

TREND REVIEW

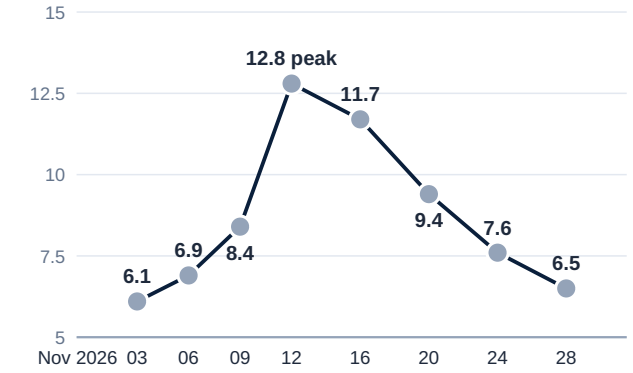


TNF-α. 8 → 10 → 15 → 24 → 20 → 15 → 11 → 9 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 24 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +1 pg/mL (+12.5%). No approved reference criterion applies, so no reference band is drawn.

White blood cell count

10^9/L

TREND REVIEW

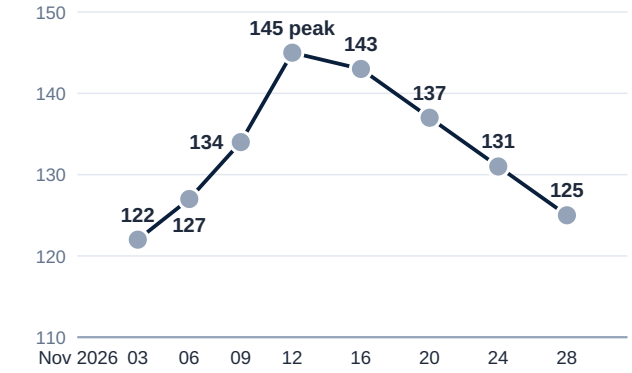


White blood cell count. 6.1 → 6.9 → 8.4 → 12.8 → 11.7 → 9.4 → 7.6 → 6.5 10^9/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 12.8 10^9/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.4 10^9/L (+6.6%). No approved reference criterion applies, so no reference band is drawn.

LDL cholesterol

mg/dL

MONITORING

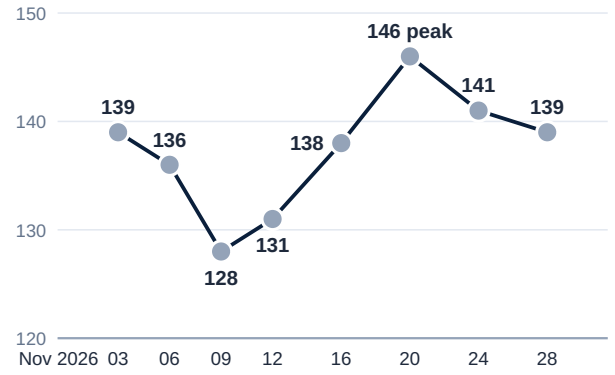


LDL cholesterol. 122 → 127 → 134 → 145 → 143 → 137 → 131 → 125 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 145 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 mg/dL (+2.5%). No approved reference criterion applies, so no reference band is drawn.

Sodium

mmol/L

MONITORING

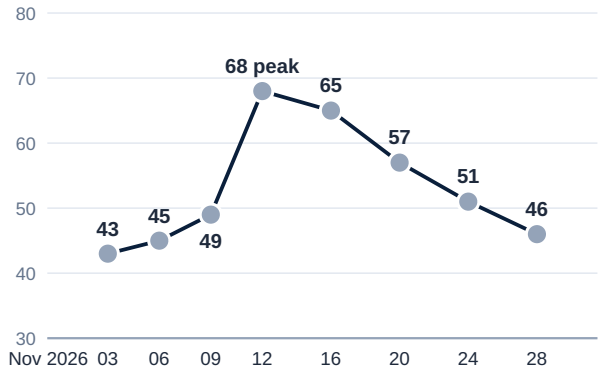


Sodium. 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 146 mmol/L on 2026-11-20 (intermediate maximum), then fell again; first to last 0 mmol/L (0%). No approved reference criterion applies, so no reference band is drawn.

P1NP (bone formation)

ng/mL

MONITORING

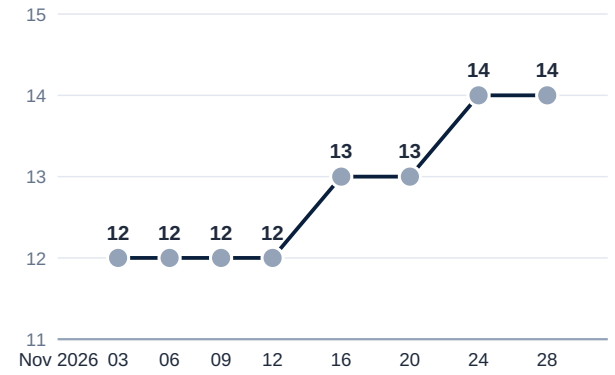


P1NP (bone formation). 43 → 45 → 49 → 68 → 65 → 57 → 51 → 46 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 68 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 ng/mL (+7%). No approved reference criterion applies, so no reference band is drawn.

Vitamin D (25-hydroxy)

ng/mL

MONITORING

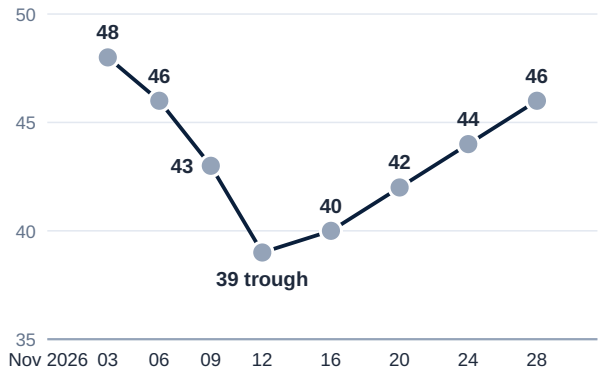


Vitamin D (25-hydroxy). 12 → 12 → 12 → 12 → 13 → 13 → 14 → 14 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Rose at every examination; first to last +2 ng/mL (+16.7%). No approved reference criterion applies, so no reference band is drawn.

HDL cholesterol

mg/dL

MONITORING



HDL cholesterol. 48 → 46 → 43 → 39 → 40 → 42 → 44 → 46 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 39 mg/dL on 2026-11-12 (intermediate minimum), then rose again; first to last -2 mg/dL (-4.2%). No approved reference criterion applies, so no reference band is drawn.

Not charted — no classification, review priority or flag in this report; every value is in the complete results table: aPTT (clotting time), Body temperature, Diastolic blood pressure, Hematocrit, Hemoglobin, Oxygen saturation (SpO2), QTc interval, Sleep efficiency, Systolic blood pressure.

Preliminary Medical Assessment — For Ground Medical Review

Automated preliminary assessment generated from the recorded measurements. It lists possible conditions for a qualified clinician to evaluate; it is not a physician-certified diagnosis and assigns no probabilities. Symptoms, examination findings, medications, age and sex were not available unless shown.

1 · Possible electrolyte disturbance (sodium / potassium)

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule WP-3). Significant change (trend review): Potassium; no RED band or diagnostic/action threshold supports high priority.
CLINICAL CLASS	• NEUTRAL — insufficient classification evidence (no applicable criterion for the contributing results).
CLINICAL CONCERN	Sodium and potassium are held within narrow limits because they control fluid distribution and the electrical activity of nerves and heart muscle. Large changes, particularly in potassium, can affect heart rhythm and therefore need prompt medical review rather than waiting for the next routine examination.
OBSERVATION	Sodium: 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 146 mmol/L on 2026-11-20 (intermediate maximum), then fell again; first to last 0 mmol/L (0%). Potassium: 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 5.8 mmol/L on 2026-11-09 (intermediate maximum), then fell again; first to last -0.1 mmol/L (-2.3%).
SUPPORTING	• Sodium: 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 146 mmol/L on 2026-11-20 (intermediate maximum), then fell again; first to last 0 mmol/L (0%). • Potassium: 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 5.8 mmol/L on 2026-11-09 (intermediate maximum), then fell again; first to last -0.1 mmol/L (-2.3%). Meets the review-triage rules.
CONTRADICTING	• Sodium fell back to 139 mmol/L at the latest examination after the 146 mmol/L peak on 2026-11-20, so the peak was not sustained. • Potassium fell back to 4.2 mmol/L at the latest examination after the 5.8 mmol/L peak on 2026-11-09, so the peak was not sustained.
ALTERNATIVES	• Sample hemolysis or delayed processing, which can falsely raise measured potassium. • Contamination from an intravenous line. • Dilution or concentration of the sample. • Laboratory variation between examinations.
MISSING	• Kidney function (creatinine / eGFR) — no accepted observation in this period. • Urine sodium and potassium, urine volume — no accepted observation in this period. • Sample handling and hemolysis index — not recorded. • Fluid intake and urine output — not recorded. • ECG findings if potassium is markedly changed — not recorded.
CONFIRM	• Repeat sodium and potassium promptly under Ground Medical supervision with careful sample handling; the timing is a clinical decision. • Interpret against the performing laboratory's own reference interval — no general interval is approved in this application.
GROUND MEDICAL	Pending Ground Medical review (session review — not persisted)

2 · Possible inflammatory response

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule WP-3). Significant change (trend review): hsCRP (inflammation), IL-6 (inflammatory signal); no RED band or diagnostic/action threshold supports high priority.
CLINICAL CLASS	• NEUTRAL — insufficient classification evidence (no applicable criterion for the contributing results).
CLINICAL CONCERN	Inflammatory markers rise when the immune system responds to infection, injury, strenuous exercise or stress. IL-6 is a short-lived signalling molecule; CRP is produced by the liver in response to IL-6 over the following day. A marked rise calls for a search for its cause.
OBSERVATION	hsCRP (inflammation): 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 28 mg/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.9 mg/L (+75%). IL-6 (inflammatory signal): 5.8 → 12 → 32 → 74 → 55 → 29 → 15 → 8.1 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 74 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +2.3 pg/mL (+39.7%).

2 · Possible inflammatory response (continued)

MONITORING OR FURTHER EVALUATION

SUPPORTING	• hsCRP (inflammation): 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 28 mg/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.9 mg/L (+75%). Meets the review-triage rules. • IL-6 (inflammatory signal): 5.8 → 12 → 32 → 74 → 55 → 29 → 15 → 8.1 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 74 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +2.3 pg/mL (+39.7%). Meets the review-triage rules.
CONTRADICTING	• hsCRP (inflammation) fell back to 2.1 mg/L at the latest examination after the 28 mg/L peak on 2026-11-12, so the peak was not sustained. • IL-6 (inflammatory signal) fell back to 8.1 pg/mL at the latest examination after the 74 pg/mL peak on 2026-11-12, so the peak was not sustained.
ALTERNATIVES	• Intense exercise before sampling. • Sample handling (IL-6 can change with processing delay). • Assay differences between examinations.
MISSING	• White-cell count and differential — no accepted observation in this period. • Body temperature and clinical examination — no accepted observation in this period. • Symptoms and body temperature at the time of sampling — not recorded. • Exercise in the preceding 24–48 hours — not recorded. • Recent injury — not recorded.
CONFIRM	• Repeat the inflammatory markers and examine for sources of infection or inflammation; the CDC/AHA statement recommends repeating hs-CRP above 10 mg/L.
GROUND MEDICAL	Pending Ground Medical review (session review — not persisted)

3 · Possible disturbance of blood-sugar regulation (dysglycemia)

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule WP-3). Significant change (trend review): Fasting glucose; no RED band or diagnostic/action threshold supports high priority. Verify first: HbA1c (3-month average blood sugar). A DATA CHECK also applies; it does not lower this priority.
CLINICAL CLASS	• NEUTRAL — insufficient classification evidence (no applicable criterion for the contributing results).
CLINICAL CONCERN	HbA1c (glycated hemoglobin) reflects average blood glucose over roughly the previous two to three months, while plasma glucose reflects the moment of sampling. Persistently raised values indicate that glucose regulation by insulin may be impaired, which over time affects blood vessels, nerves, kidneys and the eyes.
OBSERVATION	HbA1c (3-month average blood sugar): 5.8 → 5.9 → 9.1 → 5.9 → 5.8 → 5.8 → 5.8 → 5.8 % from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 9.1 % on 2026-11-09 (intermediate maximum), then fell again; first to last 0 % (0%). Fasting glucose: 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 172 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +2 mg/dL (+2.1%).
SUPPORTING	• Fasting glucose: 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 172 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +2 mg/dL (+2.1%). Meets the review-triage rules.
CONTRADICTING	• HbA1c (3-month average blood sugar): 5.9 → 9.1 % (+54.2%) from 2026-11-06 to 2026-11-09: a 3-day interval is far shorter than the period HbA1c reflects; measurement consistency must be checked before interpretation. The readings contradict each other until sample identity and method are confirmed. • HbA1c (3-month average blood sugar) fell back to 5.8 % at the latest examination after the 9.1 % peak on 2026-11-09, so the peak was not sustained. • Fasting glucose fell back to 96 mg/dL at the latest examination after the 172 mg/dL peak on 2026-11-12, so the peak was not sustained.
ALTERNATIVES	• Sample mix-up or data-entry error, especially when HbA1c changes markedly within days. • Conditions altering red-cell turnover (anemia, blood loss, transfusion, hemolysis) or hemoglobin variants affecting HbA1c. • A non-fasting or random glucose interpreted as fasting.
MISSING	• Fasting state and time of sampling — not recorded. • Recent meals, illness, stress or sleep loss — not recorded. • Sample identity and laboratory method — not recorded.
CONFIRM	• Confirm any criterion-level result with a second abnormal test (same test at a later time point, or a different test), as required by the ADA Standards of Care. • Check sample identity and laboratory method before interpreting an HbA1c change over a short interval.

3 · Possible disturbance of blood-sugar regulation (dysglycemia) (continued)

MONITORING OR FURTHER EVALUATION

GROUND MEDICAL	Pending Ground Medical review (session review — not persisted)

4 · Raised blood triglycerides (possible lipid-metabolism concern)

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule WP-3). Significant change (trend review): Triglycerides; no RED band or diagnostic/action threshold supports high priority.
CLINICAL CLASS	• NEUTRAL — insufficient classification evidence (no applicable criterion for the contributing results).
CLINICAL CONCERN	Triglycerides are the main form in which the body transports and stores fat. Levels rise for several hours after a meal; persistently raised fasting levels are associated with heart and blood-vessel complications, and very high levels can inflame the pancreas.
OBSERVATION	Triglycerides: 155 → 165 → 230 → 330 → 288 → 216 → 184 → 160 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 330 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +5 mg/dL (+3.2%). LDL cholesterol: 122 → 127 → 134 → 145 → 143 → 137 → 131 → 125 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 145 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 mg/dL (+2.5%). HDL cholesterol: 48 → 46 → 43 → 39 → 40 → 42 → 44 → 46 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 39 mg/dL on 2026-11-12 (intermediate minimum), then rose again; first to last -2 mg/dL (-4.2%).
SUPPORTING	• Triglycerides: 155 → 165 → 230 → 330 → 288 → 216 → 184 → 160 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 330 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +5 mg/dL (+3.2%). Meets the review-triage rules. • LDL cholesterol: 122 → 127 → 134 → 145 → 143 → 137 → 131 → 125 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 145 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 mg/dL (+2.5%). • HDL cholesterol: 48 → 46 → 43 → 39 → 40 → 42 → 44 → 46 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 39 mg/dL on 2026-11-12 (intermediate minimum), then rose again; first to last -2 mg/dL (-4.2%).
CONTRADICTING	• Triglycerides fell back to 160 mg/dL at the latest examination after the 330 mg/dL peak on 2026-11-12, so the peak was not sustained. • LDL cholesterol fell back to 125 mg/dL at the latest examination after the 145 mg/dL peak on 2026-11-12, so the peak was not sustained. • HDL cholesterol rose again to 46 mg/dL at the latest examination after the 39 mg/dL trough on 2026-11-12, so the trough was not sustained.
ALTERNATIVES	• Non-fasting sample (the most common cause of a single high value). • Recent alcohol intake. • Medications or other metabolic conditions affecting lipids. • Laboratory or sample-handling variation.
MISSING	• Fasting lipid panel (LDL-C, HDL-C, triglycerides) with documented fasting time — no accepted observation in this period. • ApoB — no accepted observation in this period. • Fasting state and time of the last meal — not recorded. • Recent alcohol intake — not recorded. • Current medications — not recorded.
CONFIRM	• Repeat triglycerides in a documented fasting sample before any category is applied; NHLBI bases the diagnosis on fasting levels that are regularly elevated.
GROUND MEDICAL	Pending Ground Medical review (session review — not persisted)

5 · Change in bone-turnover markers (possible bone-health concern)

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule WP-3). Significant change (trend review): CTX (bone breakdown), Osteocalcin (bone formation); no RED band or diagnostic/action threshold supports high priority.
CLINICAL CLASS	• NEUTRAL — insufficient classification evidence (no applicable criterion for the contributing results).

5 · Change in bone-turnover markers (possible bone-health concern) (continued)

MONITORING OR FURTHER EVALUATION

CLINICAL CONCERN	Bone is continuously rebuilt: osteoblasts form new bone (formation markers such as osteocalcin and P1NP) while osteoclasts break it down (resorption markers such as CTX). In microgravity, unloading increases resorption, so formation and resorption markers, calcium, parathyroid hormone and vitamin D are interpreted together.
OBSERVATION	Osteocalcin (bone formation): 3.5 → 4.1 → 5.2 → 31 → 25 → 18 → 11 → 7.2 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 31 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3.7 ng/mL (+105.7%). CTX (bone breakdown): 0.32 → 0.35 → 0.41 → 0.73 → 0.66 → 0.55 → 0.45 → 0.36 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 0.73 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +0.04 ng/mL (+12.5%). P1NP (bone formation): 43 → 45 → 49 → 68 → 65 → 57 → 51 → 46 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 68 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 ng/mL (+7%). Vitamin D (25-hydroxy): 12 → 12 → 12 → 12 → 13 → 13 → 14 → 14 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Rose at every examination; first to last +2 ng/mL (+16.7%).
SUPPORTING	• Osteocalcin (bone formation): 3.5 → 4.1 → 5.2 → 31 → 25 → 18 → 11 → 7.2 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 31 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3.7 ng/mL (+105.7%). Meets the review-triage rules. • CTX (bone breakdown): 0.32 → 0.35 → 0.41 → 0.73 → 0.66 → 0.55 → 0.45 → 0.36 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 0.73 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +0.04 ng/mL (+12.5%). Meets the review-triage rules. • P1NP (bone formation): 43 → 45 → 49 → 68 → 65 → 57 → 51 → 46 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 68 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 ng/mL (+7%). • Vitamin D (25-hydroxy): 12 → 12 → 12 → 12 → 13 → 13 → 14 → 14 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Rose at every examination; first to last +2 ng/mL (+16.7%).
CONTRADICTING	• Osteocalcin (bone formation) fell back to 7.2 ng/mL at the latest examination after the 31 ng/mL peak on 2026-11-12, so the peak was not sustained. • CTX (bone breakdown) fell back to 0.36 ng/mL at the latest examination after the 0.73 ng/mL peak on 2026-11-12, so the peak was not sustained. • P1NP (bone formation) fell back to 46 ng/mL at the latest examination after the 68 ng/mL peak on 2026-11-12, so the peak was not sustained.
ALTERNATIVES	• Assay or laboratory differences (osteocalcin is assay-dependent). • Circadian and fasting effects on bone markers. • Recent fracture healing or intense loading.
MISSING	• Parathyroid hormone (PTH) — no accepted observation in this period. • Bone density imaging (DXA / QCT) — no accepted observation in this period. • Time of day and fasting state of the bone-marker sample — not recorded. • Resistance-exercise adherence — not recorded. • Vitamin D and calcium intake — not recorded.
CONFIRM	• Measure formation and resorption markers together under the same sampling conditions; confirm a bone-health concern only with imaging and clinical evaluation.
GROUND MEDICAL	Pending Ground Medical review (session review — not persisted)

6 · Reduced heart-rate variability (possible recovery or autonomic strain)

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule WP-3). Significant change (trend review): Heart-rate variability (RMSSD); no RED band or diagnostic/action threshold supports high priority.
CLINICAL CLASS	• NEUTRAL — insufficient classification evidence (no applicable criterion for the contributing results).
CLINICAL CONCERN	Heart-rate variability (for example RMSSD) reflects the balance of the autonomic nervous system. Lower values often accompany fatigue, stress, sleep loss, illness or heavy training, and can indicate incomplete recovery; they are highly dependent on measurement conditions.
OBSERVATION	Heart-rate variability (RMSSD): 32 → 29 → 24 → 12 → 14 → 19 → 25 → 30 ms from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 12 ms on 2026-11-12 (intermediate minimum), then rose again; first to last -2 ms (-6.3%).

6 · Reduced heart-rate variability (possible recovery or autonomic strain) (continued)

MONITORING OR FURTHER EVALUATION

SUPPORTING	• Heart-rate variability (RMSSD): 32 → 29 → 24 → 12 → 14 → 19 → 25 → 30 ms from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 12 ms on 2026-11-12 (intermediate minimum), then rose again; first to last -2 ms (-6.3%). Meets the review-triage rules.
CONTRADICTING	• Heart-rate variability (RMSSD) rose again to 30 ms at the latest examination after the 12 ms trough on 2026-11-12, so the trough was not sustained.
ALTERNATIVES	• Different measurement posture, time of day or device between examinations. • Recent caffeine, exercise or meals before the recording. • Short recording duration.
MISSING	• Sleep duration and efficiency for the same nights — no accepted observation in this period. • Resting heart rate under the same conditions — no accepted observation in this period. • Recording protocol (posture, time of day, duration, device) — not recorded. • Sleep and workload in the preceding days — not recorded.
CONFIRM	• Repeat the recording under a standardised protocol and compare with the crew member's own baseline.
GROUND MEDICAL	Pending Ground Medical review (session review — not persisted)

Personalised recommendations

Each recommendation is linked to the finding that generated it. GM marks actions that require a Ground Medical decision. No medication or supplement doses are given.

Nutrition

- Record meal timing and content before blood tests. Diets built on vegetables, legumes, whole grains and adequate protein, with fewer refined sugars and sugary drinks, are generally associated with steadier blood glucose; individual changes should follow Ground Medical advice.
  → Possible disturbance of blood-sugar regulation (dysglycemia): HbA1c (3-month average blood sugar) 5.8 % on 2026-11-28; Fasting glucose …
- Triglycerides respond strongly to recent sugars, refined carbohydrates and alcohol; recording intake in the 24 hours before sampling helps interpret the result. Dietary changes should be agreed with Ground Medical.
  → Raised blood triglycerides (possible lipid-metabolism concern): Triglycerides 160 mg/dL on 2026-11-28; LDL cholesterol 125 mg/dL …
- GM Ground Medical should review calcium and vitamin D adequacy; any decision about supplementation belongs to Ground Medical.
  → Change in bone-turnover markers (possible bone-health concern): Osteocalcin (bone formation) 7.2 ng/mL on 2026-11-28; CTX …
- GM Do not change fluid or salt intake on the basis of these results until Ground Medical has reviewed them; the appropriate direction depends on the cause.
  → Possible electrolyte disturbance (sodium / potassium): Sodium 139 mmol/L on 2026-11-28; Potassium 4.2 mmol/L on 2026-11-28

Exercise

- Maintain the prescribed aerobic and resistance exercise; muscle activity increases glucose uptake and insulin sensitivity.
  → Possible disturbance of blood-sugar regulation (dysglycemia): HbA1c (3-month average blood sugar) 5.8 % on 2026-11-28; Fasting glucose …
- Regular aerobic exercise increases the clearance of triglycerides from the blood; maintain the prescribed exercise programme.
  → Raised blood triglycerides (possible lipid-metabolism concern): Triglycerides 160 mg/dL on 2026-11-28; LDL cholesterol 125 mg/dL …
- Keep to the prescribed resistance-exercise loading; mechanical loading is the main countermeasure against bone breakdown in microgravity.
  → Change in bone-turnover markers (possible bone-health concern): Osteocalcin (bone formation) 7.2 ng/mL on 2026-11-28; CTX …

Sleep and recovery

- Protect the scheduled sleep period and record sleep duration around the next recordings.
  → Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 30 ms on 2026-11-28
- Record exercise intensity, sleep and any illness symptoms in the days before the next blood test so the inflammatory markers can be placed in context.
  → Possible inflammatory response: hsCRP (inflammation) 2.1 mg/L on 2026-11-28; IL-6 (inflammatory signal) 8.1 pg/mL on 2026-11-28
- Review training load and rest days with the exercise team when heart-rate variability falls.
  → Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 30 ms on 2026-11-28
- Record heart-rate variability under the same posture, time of day and device so that changes can be compared.
  → Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 30 ms on 2026-11-28
- Repeat the changed measurements in red blood cells and blood volume with the same laboratory method at the next scheduled examination, so the change can be confirmed.
  → Red blood cells and blood volume (change below the review rules)

Medical follow-up

- GM Request a repeat HbA1c from a new sample and a fasting plasma glucose with documented fasting time, so that a criterion-level result can be confirmed or excluded.
  → Possible disturbance of blood-sugar regulation (dysglycemia): HbA1c (3-month average blood sugar) 5.8 % on 2026-11-28; Fasting glucose …
- GM Repeat a fasting lipid panel with documented fasting time to establish whether triglycerides remain raised.
  → Raised blood triglycerides (possible lipid-metabolism concern): Triglycerides 160 mg/dL on 2026-11-28; LDL cholesterol 125 mg/dL …
- GM Request formation and resorption markers together (P1NP and CTX) with PTH, and consider bone-density imaging when markers continue to change.
  → Change in bone-turnover markers (possible bone-health concern): Osteocalcin (bone formation) 7.2 ng/mL on 2026-11-28; CTX …
- GM Check sample collection and handling, fluid intake and urine output, and kidney function (creatinine or eGFR) alongside the repeat measurements.
  → Fluids, electrolytes and kidney function — Potassium (review flag)
- GM Ground Medical should review the sodium and potassium changes promptly and decide on repeat measurements; do not defer this to the next routine examination.
  → Possible electrolyte disturbance (sodium / potassium): Sodium 139 mmol/L on 2026-11-28; Potassium 4.2 mmol/L on 2026-11-28
- GM Ground Medical should review symptoms, temperature and sample conditions and decide whether repeat inflammatory markers and an examination for infection or inflammation are needed.
  → Possible inflammatory response: hsCRP (inflammation) 2.1 mg/L on 2026-11-28; IL-6 (inflammatory signal) 8.1 pg/mL on 2026-11-28
- GM Ground Medical should review the cardiovascular change and confirm that posture, timing and method were the same for both measurements.
  → Heart and circulation — Resting heart rate, Respiratory rate and Lactate (review flag)
- GM Ground Medical should review the clotting-marker change together with the venous ultrasound findings.
  → Blood clotting and venous flow — D-dimer and Fibrinogen (review flag)

Complete results — all observations

248 observations in 31 biomarker series from 8 examination date(s), exactly as recorded (no unit conversion). Classes use approved criteria. Priority is the medical-review priority of the finding the result belongs to (the same label as on page 1, in the charts and in the assessment); "Associated with HIGH PRIORITY pattern" marks a result whose own class is not high but which belongs to a high-priority pattern; Trend / data shows a significant change between examinations (TREND REVIEW) and observation-level data-quality checks (CHECK) separately.

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
FLUIDS & ELECTROLYTES						
Potassium	**4.3 mmol/L**	11/03	NEUTRAL	MONITORING	TREND REVIEW	—
Potassium	**4.8 mmol/L**	11/06	NEUTRAL	MONITORING	TREND REVIEW	—
Potassium	**5.8 mmol/L**	11/09	NEUTRAL	MONITORING	TREND REVIEW	—
Potassium	**5.3 mmol/L**	11/12	NEUTRAL	MONITORING	TREND REVIEW	—
Potassium	**4.1 mmol/L**	11/16	NEUTRAL	MONITORING	TREND REVIEW	—
Potassium	**3.2 mmol/L**	11/20	NEUTRAL	MONITORING	TREND REVIEW	—
Potassium	**3.8 mmol/L**	11/24	NEUTRAL	MONITORING	TREND REVIEW	—
Potassium	**4.2 mmol/L**	11/28	NEUTRAL	MONITORING	TREND REVIEW	—
Sodium	**139 mmol/L**	11/03	NEUTRAL	MONITORING	—	—

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
Sodium	**136 mmol/L**	11/06	NEUTRAL	MONITORING	—	—
Sodium	**128 mmol/L**	11/09	NEUTRAL	MONITORING	—	—
Sodium	**131 mmol/L**	11/12	NEUTRAL	MONITORING	—	—
Sodium	**138 mmol/L**	11/16	NEUTRAL	MONITORING	—	—
Sodium	**146 mmol/L**	11/20	NEUTRAL	MONITORING	—	—
Sodium	**141 mmol/L**	11/24	NEUTRAL	MONITORING	—	—
Sodium	**139 mmol/L**	11/28	NEUTRAL	MONITORING	—	—
GLUCOSE & LIPIDS						
Fasting glucose	**94 mg/dL**	11/03	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Fasting glucose	**108 mg/dL**	11/06	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Fasting glucose	**136 mg/dL**	11/09	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Fasting glucose	**172 mg/dL**	11/12	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Fasting glucose	**158 mg/dL**	11/16	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Fasting glucose	**119 mg/dL**	11/20	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Fasting glucose	**103 mg/dL**	11/24	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Fasting glucose	**96 mg/dL**	11/28	NEUTRAL	MONITORING	TREND REVIEW	— ◊
HbA1c (3-month average blood sugar)	**5.8 %**	11/03	NEUTRAL	MONITORING	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	**5.9 %**	11/06	NEUTRAL	MONITORING	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	**9.1 %**	11/09	NEUTRAL	MONITORING	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	**5.9 %**	11/12	NEUTRAL	MONITORING	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	**5.8 %**	11/16	NEUTRAL	MONITORING	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	**5.8 %**	11/20	NEUTRAL	MONITORING	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	**5.8 %**	11/24	NEUTRAL	MONITORING	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	**5.8 %**	11/28	NEUTRAL	MONITORING	TREND REVIEW CHECK	Held — verify the measurement first
HDL cholesterol	**48 mg/dL**	11/03	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**46 mg/dL**	11/06	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**43 mg/dL**	11/09	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**39 mg/dL**	11/12	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**40 mg/dL**	11/16	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**42 mg/dL**	11/20	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**44 mg/dL**	11/24	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**46 mg/dL**	11/28	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**122 mg/dL**	11/03	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**127 mg/dL**	11/06	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**134 mg/dL**	11/09	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**145 mg/dL**	11/12	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**143 mg/dL**	11/16	NEUTRAL	MONITORING	—	— ◊

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
LDL cholesterol	**137 mg/dL**	11/20	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**131 mg/dL**	11/24	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**125 mg/dL**	11/28	NEUTRAL	MONITORING	—	— ◊
Triglycerides	**155 mg/dL**	11/03	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
Triglycerides	**165 mg/dL**	11/06	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
Triglycerides	**230 mg/dL**	11/09	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
Triglycerides	**330 mg/dL**	11/12	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
Triglycerides	**288 mg/dL**	11/16	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
Triglycerides	**216 mg/dL**	11/20	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
Triglycerides	**184 mg/dL**	11/24	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
Triglycerides	**160 mg/dL**	11/28	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
INFLAMMATION & IMMUNITY						
hsCRP (inflammation)	**1.2 mg/L**	11/03	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**2.8 mg/L**	11/06	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**8.4 mg/L**	11/09	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**28 mg/L**	11/12	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**22 mg/L**	11/16	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**12 mg/L**	11/20	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**5.2 mg/L**	11/24	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**2.1 mg/L**	11/28	NEUTRAL	MONITORING	TREND REVIEW	Candidate criterion pending approval
IL-10 (anti-inflammatory signal)	**4 pg/mL**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**5 pg/mL**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**8 pg/mL**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**15 pg/mL**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**13 pg/mL**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**9 pg/mL**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**6 pg/mL**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**5 pg/mL**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
IL-6 (inflammatory signal)	**5.8 pg/mL**	11/03	NEUTRAL	MONITORING	TREND REVIEW	—
IL-6 (inflammatory signal)	**12 pg/mL**	11/06	NEUTRAL	MONITORING	TREND REVIEW	—
IL-6 (inflammatory signal)	**32 pg/mL**	11/09	NEUTRAL	MONITORING	TREND REVIEW	—
IL-6 (inflammatory signal)	**74 pg/mL**	11/12	NEUTRAL	MONITORING	TREND REVIEW	—
IL-6 (inflammatory signal)	**55 pg/mL**	11/16	NEUTRAL	MONITORING	TREND REVIEW	—
IL-6 (inflammatory signal)	**29 pg/mL**	11/20	NEUTRAL	MONITORING	TREND REVIEW	—
IL-6 (inflammatory signal)	**15 pg/mL**	11/24	NEUTRAL	MONITORING	TREND REVIEW	—
IL-6 (inflammatory signal)	**8.1 pg/mL**	11/28	NEUTRAL	MONITORING	TREND REVIEW	—
TNF-α	**8 pg/mL**	11/03	NEUTRAL	—	TREND REVIEW	— ◊

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
TNF-α	**10 pg/mL**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**15 pg/mL**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**24 pg/mL**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**20 pg/mL**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**15 pg/mL**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**11 pg/mL**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**9 pg/mL**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
BLOOD & CLOTTING						
aPTT (clotting time)	**29 s**	11/03	NEUTRAL	—	—	—
aPTT (clotting time)	**30 s**	11/06	NEUTRAL	—	—	—
aPTT (clotting time)	**31 s**	11/09	NEUTRAL	—	—	—
aPTT (clotting time)	**34 s**	11/12	NEUTRAL	—	—	—
aPTT (clotting time)	**32 s**	11/16	NEUTRAL	—	—	—
aPTT (clotting time)	**31 s**	11/20	NEUTRAL	—	—	—
aPTT (clotting time)	**30 s**	11/24	NEUTRAL	—	—	—
aPTT (clotting time)	**29 s**	11/28	NEUTRAL	—	—	—
D-dimer	**0.32 mg/L FEU**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**0.41 mg/L FEU**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**0.82 mg/L FEU**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**2.8 mg/L FEU**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**2.1 mg/L FEU**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**1.2 mg/L FEU**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**0.69 mg/L FEU**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**0.39 mg/L FEU**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**295 mg/dL**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**320 mg/dL**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**390 mg/dL**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**520 mg/dL**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**485 mg/dL**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**420 mg/dL**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**350 mg/dL**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**305 mg/dL**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
Hematocrit	**43 %**	11/03	NEUTRAL	—	—	—
Hematocrit	**42 %**	11/06	NEUTRAL	—	—	—
Hematocrit	**41 %**	11/09	NEUTRAL	—	—	—
Hematocrit	**38 %**	11/12	NEUTRAL	—	—	—
Hematocrit	**37 %**	11/16	NEUTRAL	—	—	—
Hematocrit	**39 %**	11/20	NEUTRAL	—	—	—
Hematocrit	**40 %**	11/24	NEUTRAL	—	—	—
Hematocrit	**42 %**	11/28	NEUTRAL	—	—	—
Hemoglobin	**14.5 g/dL**	11/03	NEUTRAL	—	—	—
Hemoglobin	**14.2 g/dL**	11/06	NEUTRAL	—	—	—
Hemoglobin	**13.8 g/dL**	11/09	NEUTRAL	—	—	—

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
Hemoglobin	**12.6 g/dL**	11/12	NEUTRAL	—	—	—
Hemoglobin	**12.4 g/dL**	11/16	NEUTRAL	—	—	—
Hemoglobin	**12.9 g/dL**	11/20	NEUTRAL	—	—	—
Hemoglobin	**13.5 g/dL**	11/24	NEUTRAL	—	—	—
Hemoglobin	**14.0 g/dL**	11/28	NEUTRAL	—	—	—
HEART & CIRCULATION						
Diastolic blood pressure	**76 mmHg**	11/03	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**73 mmHg**	11/06	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**69 mmHg**	11/09	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**54 mmHg**	11/12	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**58 mmHg**	11/16	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**65 mmHg**	11/20	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**72 mmHg**	11/24	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**75 mmHg**	11/28	NEUTRAL	—	—	— ◊
Lactate	**1.1 mmol/L**	11/03	NEUTRAL	—	TREND REVIEW	—
Lactate	**1.3 mmol/L**	11/06	NEUTRAL	—	TREND REVIEW	—
Lactate	**1.8 mmol/L**	11/09	NEUTRAL	—	TREND REVIEW	—
Lactate	**4.2 mmol/L**	11/12	NEUTRAL	—	TREND REVIEW	—
Lactate	**3.0 mmol/L**	11/16	NEUTRAL	—	TREND REVIEW	—
Lactate	**1.9 mmol/L**	11/20	NEUTRAL	—	TREND REVIEW	—
Lactate	**1.4 mmol/L**	11/24	NEUTRAL	—	TREND REVIEW	—
Lactate	**1.2 mmol/L**	11/28	NEUTRAL	—	TREND REVIEW	—
QTc interval	**416 ms**	11/03	NEUTRAL	—	—	—
QTc interval	**425 ms**	11/06	NEUTRAL	—	—	—
QTc interval	**447 ms**	11/09	NEUTRAL	—	—	—
QTc interval	**510 ms**	11/12	NEUTRAL	—	—	—
QTc interval	**484 ms**	11/16	NEUTRAL	—	—	—
QTc interval	**455 ms**	11/20	NEUTRAL	—	—	—
QTc interval	**433 ms**	11/24	NEUTRAL	—	—	—
QTc interval	**421 ms**	11/28	NEUTRAL	—	—	—
Resting heart rate	**64 bpm**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**69 bpm**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**77 bpm**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**108 bpm**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**98 bpm**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**83 bpm**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**72 bpm**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**66 bpm**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
Systolic blood pressure	**118 mmHg**	11/03	NEUTRAL	—	—	— ◊
Systolic blood pressure	**114 mmHg**	11/06	NEUTRAL	—	—	— ◊
Systolic blood pressure	**108 mmHg**	11/09	NEUTRAL	—	—	— ◊
Systolic blood pressure	**88 mmHg**	11/12	NEUTRAL	—	—	— ◊
Systolic blood pressure	**92 mmHg**	11/16	NEUTRAL	—	—	— ◊
Systolic blood pressure	**103 mmHg**	11/20	NEUTRAL	—	—	— ◊

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
Systolic blood pressure	**112 mmHg**	11/24	NEUTRAL	—	—	— ◊
Systolic blood pressure	**117 mmHg**	11/28	NEUTRAL	—	—	— ◊
BONE & MINERAL						
CTX (bone breakdown)	**0.32 ng/mL**	11/03	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.35 ng/mL**	11/06	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.41 ng/mL**	11/09	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.73 ng/mL**	11/12	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.66 ng/mL**	11/16	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.55 ng/mL**	11/20	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.45 ng/mL**	11/24	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.36 ng/mL**	11/28	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**3.5 ng/mL**	11/03	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**4.1 ng/mL**	11/06	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**5.2 ng/mL**	11/09	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**31 ng/mL**	11/12	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**25 ng/mL**	11/16	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**18 ng/mL**	11/20	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**11 ng/mL**	11/24	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**7.2 ng/mL**	11/28	NEUTRAL	MONITORING	TREND REVIEW	— ◊
P1NP (bone formation)	**43 ng/mL**	11/03	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**45 ng/mL**	11/06	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**49 ng/mL**	11/09	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**68 ng/mL**	11/12	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**65 ng/mL**	11/16	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**57 ng/mL**	11/20	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**51 ng/mL**	11/24	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**46 ng/mL**	11/28	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**12 ng/mL**	11/03	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**12 ng/mL**	11/06	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**12 ng/mL**	11/09	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**12 ng/mL**	11/12	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**13 ng/mL**	11/16	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**13 ng/mL**	11/20	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**14 ng/mL**	11/24	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**14 ng/mL**	11/28	NEUTRAL	MONITORING	—	— ◊
SLEEP & RECOVERY						
Heart-rate variability (RMSSD)	**32 ms**	11/03	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**29 ms**	11/06	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**24 ms**	11/09	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**12 ms**	11/12	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**14 ms**	11/16	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**19 ms**	11/20	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**25 ms**	11/24	NEUTRAL	MONITORING	TREND REVIEW	—

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
Heart-rate variability (RMSSD)	**30 ms**	11/28	NEUTRAL	MONITORING	TREND REVIEW	—
Sleep efficiency	**83 %**	11/03	NEUTRAL	—	—	— ◊
Sleep efficiency	**78 %**	11/06	NEUTRAL	—	—	— ◊
Sleep efficiency	**69 %**	11/09	NEUTRAL	—	—	— ◊
Sleep efficiency	**52 %**	11/12	NEUTRAL	—	—	— ◊
Sleep efficiency	**56 %**	11/16	NEUTRAL	—	—	— ◊
Sleep efficiency	**66 %**	11/20	NEUTRAL	—	—	— ◊
Sleep efficiency	**75 %**	11/24	NEUTRAL	—	—	— ◊
Sleep efficiency	**81 %**	11/28	NEUTRAL	—	—	— ◊
OTHER MEASUREMENTS						
Body temperature	**36.7 °C**	11/03	NEUTRAL	—	—	— ◊
Body temperature	**37.0 °C**	11/06	NEUTRAL	—	—	— ◊
Body temperature	**37.5 °C**	11/09	NEUTRAL	—	—	— ◊
Body temperature	**38.8 °C**	11/12	NEUTRAL	—	—	— ◊
Body temperature	**38.2 °C**	11/16	NEUTRAL	—	—	— ◊
Body temperature	**37.6 °C**	11/20	NEUTRAL	—	—	— ◊
Body temperature	**37.1 °C**	11/24	NEUTRAL	—	—	— ◊
Body temperature	**36.8 °C**	11/28	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	**98 %**	11/03	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	**97 %**	11/06	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	**96 %**	11/09	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	**91 %**	11/12	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	**93 %**	11/16	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	**96 %**	11/20	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	**97 %**	11/24	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	**98 %**	11/28	NEUTRAL	—	—	— ◊
Respiratory rate	**14 breaths/min**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	**15 breaths/min**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	**17 breaths/min**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	**24 breaths/min**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	**21 breaths/min**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	**18 breaths/min**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	**16 breaths/min**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	**14 breaths/min**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	**6.1 10^9/L**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	**6.9 10^9/L**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	**8.4 10^9/L**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	**12.8 10^9/L**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	**11.7 10^9/L**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	**9.4 10^9/L**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	**7.6 10^9/L**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	**6.5 10^9/L**	11/28	NEUTRAL	—	TREND REVIEW	— ◊

— No applicable criterion: NEUTRAL (insufficient classification evidence).

◊ Synthetic demonstration catalog identity (not scientifically validated).

Methodology note

• Data source and census. Census p60-observation-census.v1: 248 observations, 31 series, dates 2026-11-03, 2026-11-06, 2026-11-09, 2026-11-12, 2026-11-16, 2026-11-20, 2026-11-24, 2026-11-28; SHA-256 8aedfd555c01535527e1e1e60cee334b5b7b7f4e5071d281255677e314a0be59. Records held only on another device or only in the cloud are not included unless imported through the explicit cross-device sync. Generator p60-premium-report.v1 · Evidence ID MITO-EV-EDD5-8690-35DF-9594.

• Data. 248 observations · 31 biomarker series (exact catalog identity + unit) · 8 examination date(s) · 31 series compared between examinations. Only accepted, stored observations are evaluated; rows held at import for Researcher review (unresolved names or units) are not stored and are not evaluated in this report. Values exactly as recorded; no unit conversion or interpolation. Synthetic demonstration data.

• Calculations. Changes only between consecutive observations of the same identity and unit; the full trajectory, including intermediate peaks and troughs, is reported.

• Criteria. 0 Researcher-approved criteria available. Provisional literature interpretation OFF. General-population criteria are not NASA flight limits.

• Warning priority (panel colour; separate from clinical class; owner-directed, pending Researcher review). WP-1 HIGH PRIORITY MEDICAL REVIEW: A contributing result lies in a RED band of an approved criterion, or of a documented literature criterion (provisional, synthetic demonstration) whose per-measurement conditions are all recorded or that has none. WP-2 HIGH PRIORITY MEDICAL REVIEW: A contributing series shows a significant change under the trend-review rules AND a contributing result lies in a band of a diagnostic or action threshold (approved, or provisional literature with no unrecorded per-measurement condition). WP-3 MONITORING OR FURTHER EVALUATION: Significant change (trend review) without WP-1/WP-2, or a contributing result in an AMBER band of a risk category, reference interval or threshold. WP-4 DATA CHECK: Measurement-consistency check required. Shown as the panel colour only when no WP-1 to WP-3 evidence exists from other (non-held) results; otherwise added as a separate DATA CHECK badge. WP-5 WITHIN APPLICABLE REFERENCE CRITERIA: Every contributing classified result lies within a normal band and no review or data-quality flag applies. NEUTRAL: No defensible classification is available.

• Cross-domain pattern analysis (P60) runs only in the literature-based synthetic demonstration; it is not applied to real data or without provisional literature mode.

• Automated assessment. Deterministic clinical reasoning engine. Review priority and data quality are separate from clinical classification. Not a diagnosis; no medication or dose advice.